

ARTI 403 - Image Processing

Lab 2 Report

Sampling, Quantization, Arithmetic and Logical Operations

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Objective: apply image sampling and quantization, arithmetic operations (addition, subtraction, constant addition) and set/logical operations (union, set difference, symmetric difference, intersection) on grayscale and binary images using Python, OpenCV, NumPy and PIL.

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1. Lab Overview

This report documents the completed activities for Lab 2. The work was implemented in a Jupyter Notebook using Python with **OpenCV**, **NumPy**, **Matplotlib** and **PIL**.

Two grayscale test images were used for the sampling, quantization and arithmetic tasks: **lena_gray_256.tif** and **cameraman.tif**. Two binary shape images, **A.png** and **B.png**, were used for the set and logical operations. For the arithmetic and logical operations, all images were resized to 400 x 400 pixels using Lanczos interpolation so that both operands have the same size.

File	Type	Used in
lena_gray_256.tif	Grayscale, 256 x 256	Sampling, quantization, arithmetic operations
cameraman.tif	Grayscale	Addition and subtraction with Lena
A.png	Binary shape image	Set and logical operations
B.png	Binary shape image	Set and logical operations

2. Sampling and Quantization

The Lena image was downsampled with **sample_image()** (nearest-neighbour resize by an integer factor) and its gray levels were reduced with **quantize_image()**. The parameters were a sampling factor of **50** and **2** quantization levels.

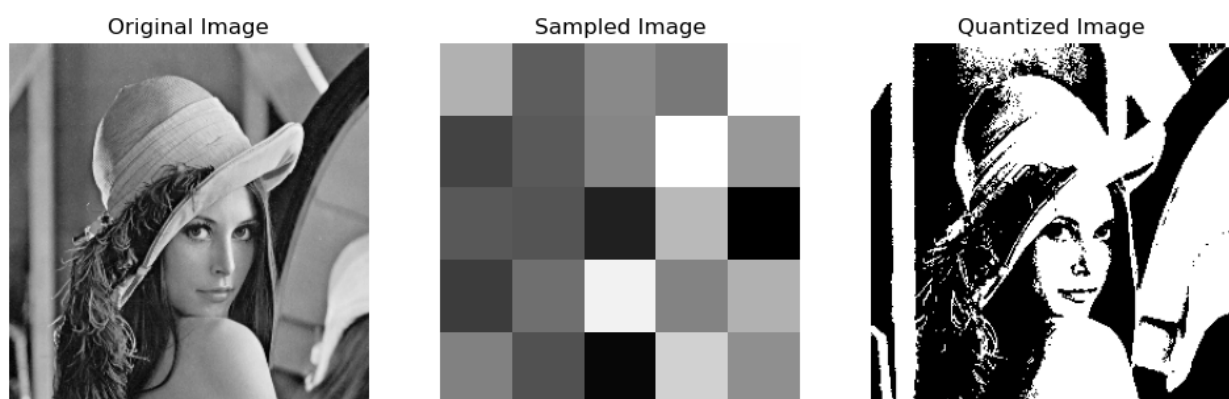


Figure 1. Original, sampled (factor 50) and quantized (2 levels) Lena image.

Observation: A factor of 50 reduces the 256 x 256 image to only 5 x 5 samples ($256 // 50$), so each sample becomes a large uniform block and almost all spatial detail is lost. With 2 levels the step is $256 // 2 = 128$, so every pixel is mapped to 0 or 128; the quantized image keeps the overall shape of the face but loses all smooth gray transitions (false contouring).

3. Arithmetic Operations

Lena and Cameraman were resized to 400 x 400 and combined pixel by pixel. The three operations below differ in how they handle values outside the 0-255 range.

3.1 Addition of Two Images

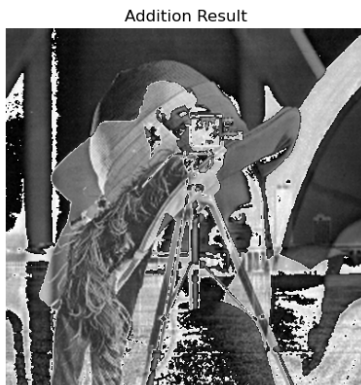


Figure 2. Lena + Cameraman.

The two images were added with NumPy's `+` operator on uint8 arrays.

```
addition = im1arr + im2arr
```

Observation: NumPy uint8 addition wraps around modulo 256 instead of saturating. Where both images are bright, the sum exceeds 255 and wraps to a small value, which creates the dark patches and noisy artifacts in the result.

3.2 Subtraction of Two Images



Figure 3. Lena - Cameraman.

Cameraman was subtracted from Lena using OpenCV's saturating subtraction.

```
cv2.subtract(im1arr, im2arr)
```

Observation: `cv2.subtract` clips negative results to 0. Only the regions where Lena is brighter than Cameraman remain visible, so the result is mostly dark and highlights the differences between the two images.

3.3 Adding a Constant Value (175)

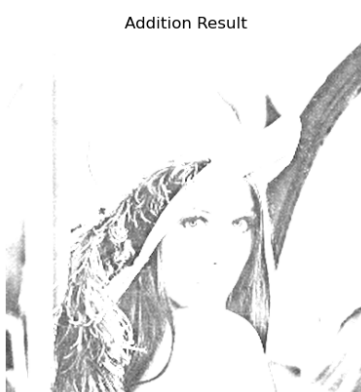


Figure 4. Lena + 175.

A constant value of 175 was added to the Lena image using OpenCV's saturating addition.

```
cv2.add(im1arr, 175)
```

Observation: `cv2.add` saturates at 255, so every pixel with intensity of 80 or more becomes white. The image is strongly brightened and washed out, and only the originally dark regions keep visible detail.

4. Set and Logical Operations

The binary images A (two horizontally overlapping circles) and B (two vertically overlapping circles) were resized to 400 x 400 and combined with bitwise operators. White pixels belong to the set and black pixels to the background.

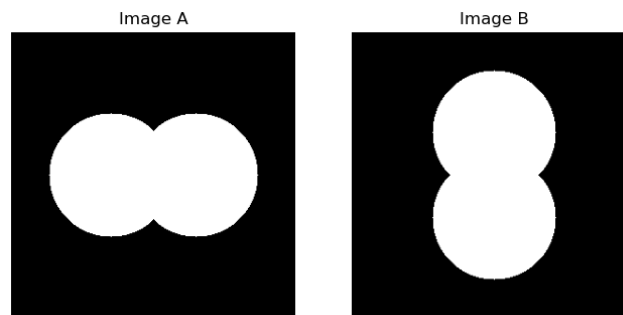


Figure 5. Input images A and B.

Operation	Set notation	Code	Figure
Union	$A \cup B$	<code>im4arr im3arr</code>	Figure 6
Set difference	$A - B$	<code>im3arr & (~im4arr)</code>	Figure 7
Symmetric difference	$A \Delta B$	<code>im3arr ^ im4arr</code>	Figure 8
Intersection	$A \cap B$	<code>im3arr & im4arr</code>	Figure 9

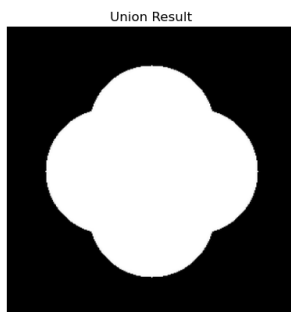


Figure 6.
Union

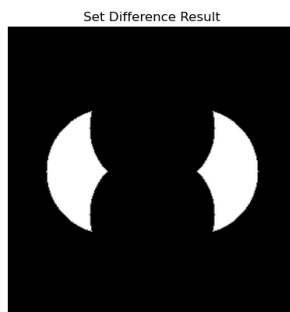


Figure 7.
Set difference

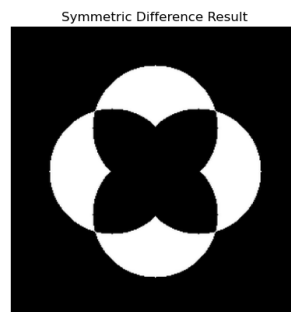


Figure 8.
Symmetric difference

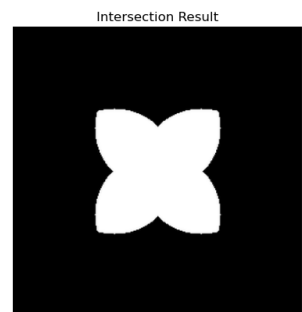


Figure 9.
Intersection

Observation: The union merges both shapes into a single larger shape. The set difference keeps only the left and right parts of A that lie outside B. The symmetric difference keeps the pixels that belong to exactly one of the two images (the union without the overlap), and the intersection keeps only the central region shared by A and B.

5. Final Results Summary

All the required operations were implemented in the notebook and their results are shown in this report.

No.	Operation	Input / Configuration	Result
1	Sampling	Lena; factor 50	5 x 5 sampled image
2	Quantization	Lena; 2 levels	Two-level image
3	Image addition	Lena + Cameraman (400 x 400)	Wrap-around artifacts
4	Image subtraction	Lena - Cameraman	Dark difference image
5	Constant addition	Lena + 175	Bright, saturated image
6	Set difference	A and B (binary)	Parts of A outside B
7	Symmetric difference	A and B (binary)	Union without overlap
8	Intersection	A and B (binary)	Region shared by A and B
9	Union	A and B (binary)	Combined shape

6. Submission Contents

The submission contains the completed Jupyter notebook **ARTI403_Lab2_2240005501.ipynb** together with the images used by it (lena_gray_256.tif, cameraman.tif, A.png and B.png).

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